Matt Baughen

MSc Design Engineering student at Imperial College London

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Experience

Heat Recovery Engineer

Oct 2024 – Sep 2026 (full time)
Thermal Energy (UK) Ltd.
Bristol, UK

Engineering analysis

- Responsible for preliminary analysis and project development for heat recovery projects using in-house modelling software
- Developed process flow diagrams (PFDs), costing estimates, budgetary proposals and presentations for a large range of multinational clients in a timely and efficient manner
- Co-authored a technical white paper published by *TÜV Rheinland* on multi-stage condensing economiser technology

Project engineering & support

Supported industrial heat recovery project teams across Europe, including time-pressured EC&I troubleshooting using Siemens TIA Portal and data analysis in Microsoft Excel for system performance verification

Business development & automation

- Led the development of a Django-based automated quotation system for boiler economiser projects, reducing engineering hours by 75-80%
- Proposed and fully developed a Python data processing tool for third-party steam trap surveys, reducing administrative workload by 50-70%
- Currently working part-time alongside my studies to develop an automated sizing engine for steam trap sizing calculations and associated admin tasks using Python machine learning & data science toolkits

Student Research Intern

Jun – Jul 2024
Centre for Climate Adaptation and Environment Research
University of Bath, UK

Utilised discrete signal processing techniques alongside wind speed readings from the ESA's Aeolus satellite and the ERA5 reanalysis dataset to infer the likelihood of gravity wave formation based on topographic and atmospheric factors using MATLAB

Software Engineering Intern

Jun – Aug 2023
3SC Ltd.
Derbyshire, UK

- Developed C# unit tests for a client's CRM project using xUnit and made front-end improvements to CRM web apps using Sass
- Provided on-site assistance to the EC&I team for a system upgrade to the Heathrow T5 pod system

Education

Imperial College London

MSc Design Engineering
Since Sep 2026

University of Bath

BEng (Hons) Mechanical and Electrical Engineering
Sep 2021 – Jun 2024 | First-class honours

Skills

Technical writing & research
LaTeX/Overleaf, Zotero, Markdown, Microsoft Word/PowerPoint

Web development
Linux, HTML, CSS/Sass, JavaScript, Django/DRF, Streamlit

Rapid prototyping
CAD (Autodesk Fusion, KiCAD), FDM 3D printing (Prusa/Bambu), embedded systems (C++, STM32/ESP32, Arduino, Raspberry Pi)

Data analysis & automation
Python (NumPy, Pandas), MATLAB, Siemens TIA Portal, Microsoft Excel

Languages
English (native), German (lower intermediate)

Additional information

Student Sustainability Ambassador
Imperial College London, Since Oct 2026

Member, Institution of Engineering and Technology (MIET)
Since Apr 2025

Full UK driving licence
Since Aug 2021

References available upon request